



LapRF Personal Timing System
Preliminary Operator's Manual

Rev 0.99 - May 2017



IMMERSION **RC**
REAL VIRTUALITY

Specifications

5.8GHz RF Detection	
Frequency	5.3-6.0GHz
Programmable Attenuation	0-32dB
Antenna	Custom linearly polarized (Suitable for LHCP, RHCP, Linear use)
Channels	Arbitrary (defined by connected app.)
Pilots	1-8, depending upon required precision and gate speed
Mechanical	
Dimensions	L = 100mm, W = 100mm, H = 25mm
Weight	121g
Battery	
Cell	18650, 2600mAh, 3.7V nominal
Consumption	~300mA @ 3.7V
Bluetooth	
Standard	BLE, Bluetooth Smart
Antenna	Internal CE-compliant, or external dipole for extra range (up to 50m)
App Support	
Platforms	iOS and Android
Software Interface	
Ports	Bluetooth Wireless, and USB Wired
Protocol	LapRF Comms Protocol (available upon request for developers)
PC Software	LiveTime scoring engine <i>late June 2017</i>

Introduction

The LapRF Personal Edition is the first in a family of race timing systems from ImmersionRC.

LapRF technology uses the emitted 5.8GHz RF from race quadcopters (or fixed wing aircraft) to detect a passage through the start-finish gate.

Different from other offerings on the market, the LapRF uses custom timing receivers with specially designed antennas to ensure that only quads which pass through the 'beam' are timed, and other sources of interference are ignored.

All that is required to use the LapRF in a race is to program the band, and channel that each pilot will use, place the timer in the start/finish gate, and go fly! No transponders required.

Getting Started

Charging the Battery

Using a standard USB Micro cable charge the timing puck for at least 3 hours before use.

Ok, so nobody actually wants to do this... the battery is delivered approx. 70% charged, and won't be damaged if the timer is powered up, and a few laps flown. Do not run it for long though before charging it to prevent permanent damage to the battery.

App. Download

Search for the term 'LapRF' in either the iOS App. store, or the Google play store.

Download the free app. and install.

Powering On

To power on the LapRF, press and hold the power button, located on the underside of the timer, for about 4 seconds until the LEDs light up, and the LapRF beeps.

To power off, repeat this procedure.

Connecting to the LapRF

iOS: Open the 'Race Setup' page of the LapRF app, and select your timer's ID (shown on the label on the bottom of the timer) from the list shown.

The 'Start Race' icon on the Race page should go green to indicate a good connection.

Configuring the Bands/Channels

Once the LapRF is connected, click on the 'LapRF Setup' icon and configure your bands and channels.

The switch on the left enables/disables each pilot slot, be sure to disable unused slots for maximum accuracy.

Once the band and channel have been selected, the frequency is displayed.

Use the power switch to select the power level that you will be running.

Click the 'Write Settings to LapRF' button to program the time. Note that once these are programmed, the settings will persist even when the timer is power cycled.

Finally, Race!

Click on the Race button, and then 'Start Race'.

'Goggles Down, Thumbs up'

Joe Scully - Some time in 2015

Placement

The lapRF's integrated patch antenna forms a beam which emits from the top of the product.

Ideal placement is in the center of the gate, but that does expose it to strikes from low-flying (high speed) racers.

A small hole in the ground works well to protect it for grass tracks, or where this is not possible, either protect the timer from strikes placing a solid object in front of it, or place the timer at a 45 degree angle from one corner of the gate.

When laying out tracks for RF timing, avoid parts of the track which come close to the timing gate to prevent false readings.

For start grid positioning, place it at least 5m from the gate to ensure that the first crossing is registered.

Battery Charging

The internal 18650 li-ion battery may be charged by simply connecting the puck to a USB power source using a standard micro-B cable.

The maximum charge current is approx. 800mA, which results in a charge time of a little more than 3 hours.

For extended use, the 18650 may be easily swapped out and replaced with a fully charged pack.



WARNING: 18650 batteries can be extremely dangerous if mishandled. Keeping unprotected batteries in your pocket, especially with other metallic objects, can result in an explosion, and severe burns.

Always use a protective carrying case when transporting these cells.

When purchasing batteries, name-brand cells from reputable sources are highly recommended. Many of the 'budget' cells available from online resellers may not be as safe as genuine cells.

Battery Life

The supplied 2600mAh cell should suffice for approx. 8 hours of racing. As the cell voltage reduces, the gate will automatically turn off, after beeping to warn the user approx. 5 minutes before.

iOS Application for iPhone and iPad

Main Screen

The main screen shows the lap times, including average, best, and total times, for each active pilot.

Times in green are the best lap times for each pilot.

Note: When a timing system is not connected, touching the large pilot number on the left side of the screen will create a manual time, and can be used to evaluate the app.

The color of the 'Start Race' button currently indicates a valid connection to the lapRF. Red for no connection, Green for a good connection.

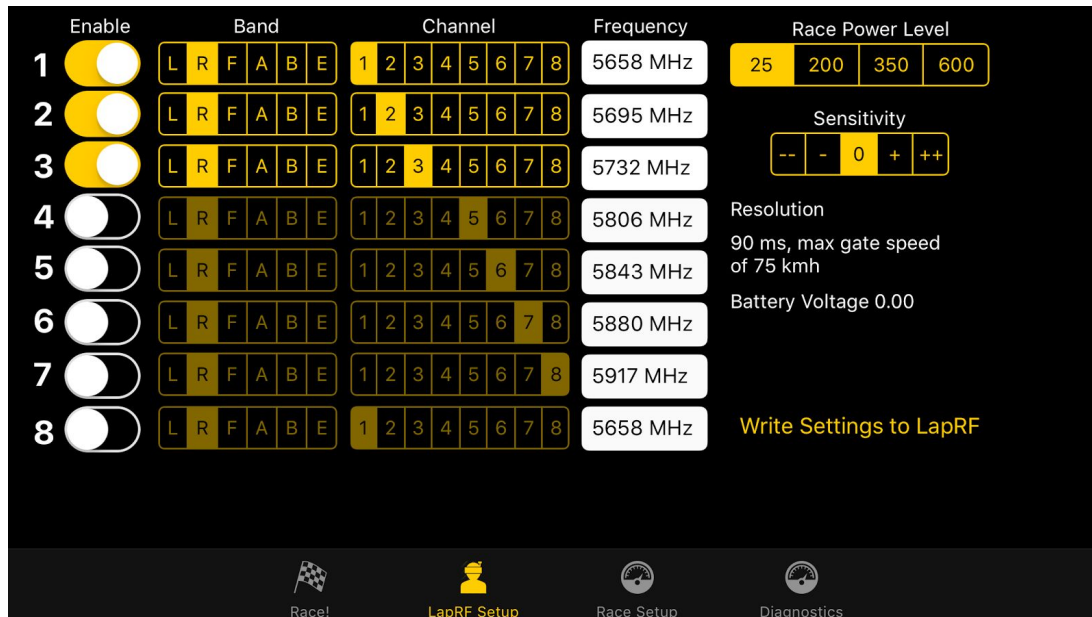
			Race Clock 069:7												
			Lap 1	Lap 2	Lap 3	Lap 4	Lap 5	Lap 6	Lap 7	Lap 8	Avg	Best	Total	#	
1	Pilot 1		2.87	19.24	1.71	-	-	-	-	-	7.94	1.71	23.82	3	
2	Pilot 2		0.97	15.32	-	-	-	-	-	-	8.14	0.97	16.29	2	
3	Pilot 3		-	-	-	-	-	-	-	-	-	-	0.00	-	
4	Pilot 4		16.24	-	-	-	-	-	-	-	16.24	16.24	16.24	1	
5	Pilot 5		1.40	12.57	-	-	-	-	-	-	6.98	1.40	13.97	2	
6	Pilot 6		16.97	3.80	-	-	-	-	-	-	10.38	3.80	20.77	2	
7	Pilot 7		-	-	-	-	-	-	-	-	-	-	0.00	-	
8	Pilot 8		-	-	-	-	-	-	-	-	-	-	0.00	-	
			 Race!	 LapRF Setup	 Race Setup	 Diagnostics									

LapRF Setup

The lapRF setup page is where the number of active pilots is set, plus the band and channel that each pilot is using.

In addition, the power level used for the race (all pilots should ideally use the same vTx power) is set here, along with a 'sensitivity tweak' which works a bit like the exposure correction on most high-end cameras.

If you are missing laps, increase sensitivity, if you are getting false hits, decrease it.



Enable	Band	Channel	Frequency	Race Power Level
1 ☒	L R F A B E	1 2 3 4 5 6 7 8	5658 MHz	25 200 350 600
2 ☒	L R F A B E	1 2 3 4 5 6 7 8	5695 MHz	
3 ☒	L R F A B E	1 2 3 4 5 6 7 8	5732 MHz	
4 ☐	L R F A B E	1 2 3 4 5 6 7 8	5806 MHz	
5 ☐	L R F A B E	1 2 3 4 5 6 7 8	5843 MHz	
6 ☐	L R F A B E	1 2 3 4 5 6 7 8	5880 MHz	
7 ☐	L R F A B E	1 2 3 4 5 6 7 8	5917 MHz	
8 ☐	L R F A B E	1 2 3 4 5 6 7 8	5658 MHz	

Sensitivity: -- - 0 + ++

Resolution: 90 ms, max gate speed of 75 kmh
Battery Voltage: 0.00

Write Settings to LapRF

Navigation: Race! LapRF Setup Race Setup Diagnostics

This is a good time to talk about pilot count, and its effect on resolution/accuracy/gate speed.

The personal time has a single timing module, which scans each of the active channels.

When a single pilot is enabled, the RF signal is 'sampled' at the maximum rate, and supports quads flying through the start/finish gate at over 160km/h.

When a second pilot is enabled, the timing module has to toggle between the two channels, reducing the accuracy of each channel, and decreasing maximum gate speed.

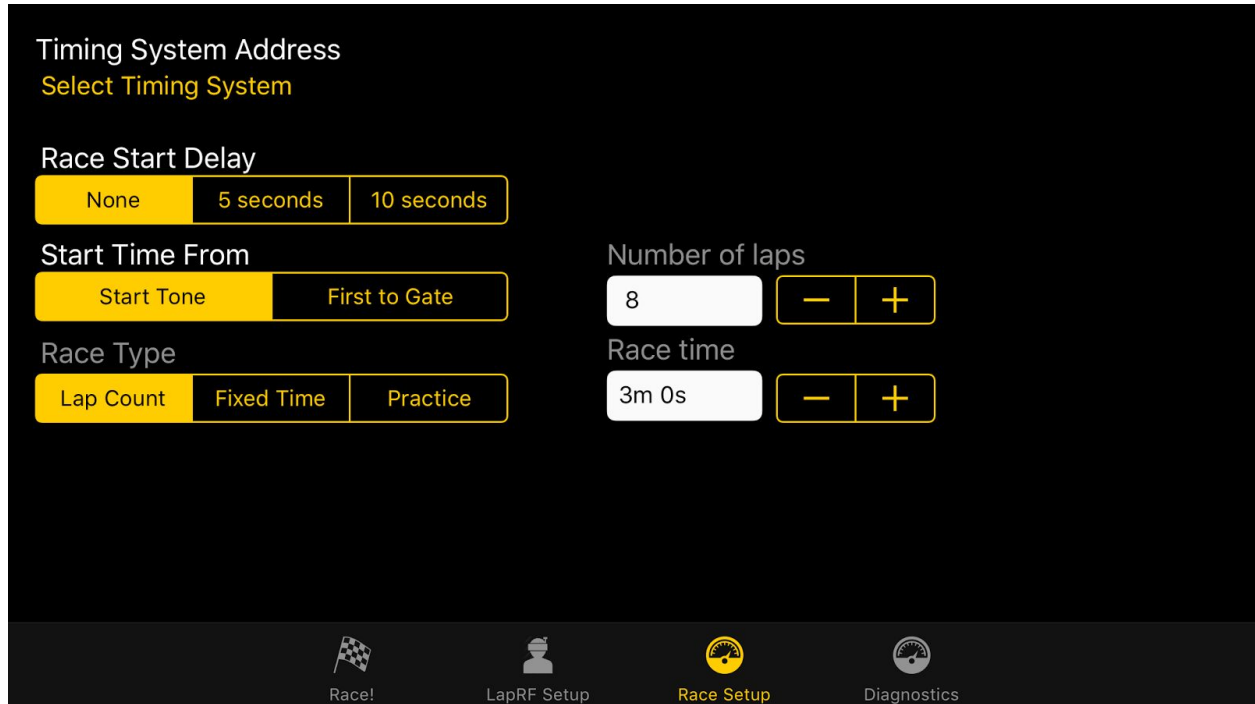
In practice, for 250 sized quads, with pilots of typical skill level, 4 pilots can easily share a single lap timer.

For slower quads, tiny-whoop class for example, even more pilot slots may be enabled.

For the best accuracy at 200km/h+ gate speeds, and up to 16 pilots, the lapRF Event system is a better choice, and should be used for larger events. *Availability July 2017.*

Race Setup

The Race Setup page is where miscellaneous race settings/preferences are configured.



Race Start Delay

Configure the delay between hitting the 'Start Race' button and the race starting. Use it to give time to get your goggles on before flying.

Start Time From

FPV races are usually either configured to start from the tone (start grid is positioned after the Start/Finish gate), or from 'First to Gate', where the start grid is positioned before it.

With the latter, lap times are for full laps, including the first lap.

Race Type

For races with a fixed number of laps, select 'Lap Count', and set the number of laps to be raced.

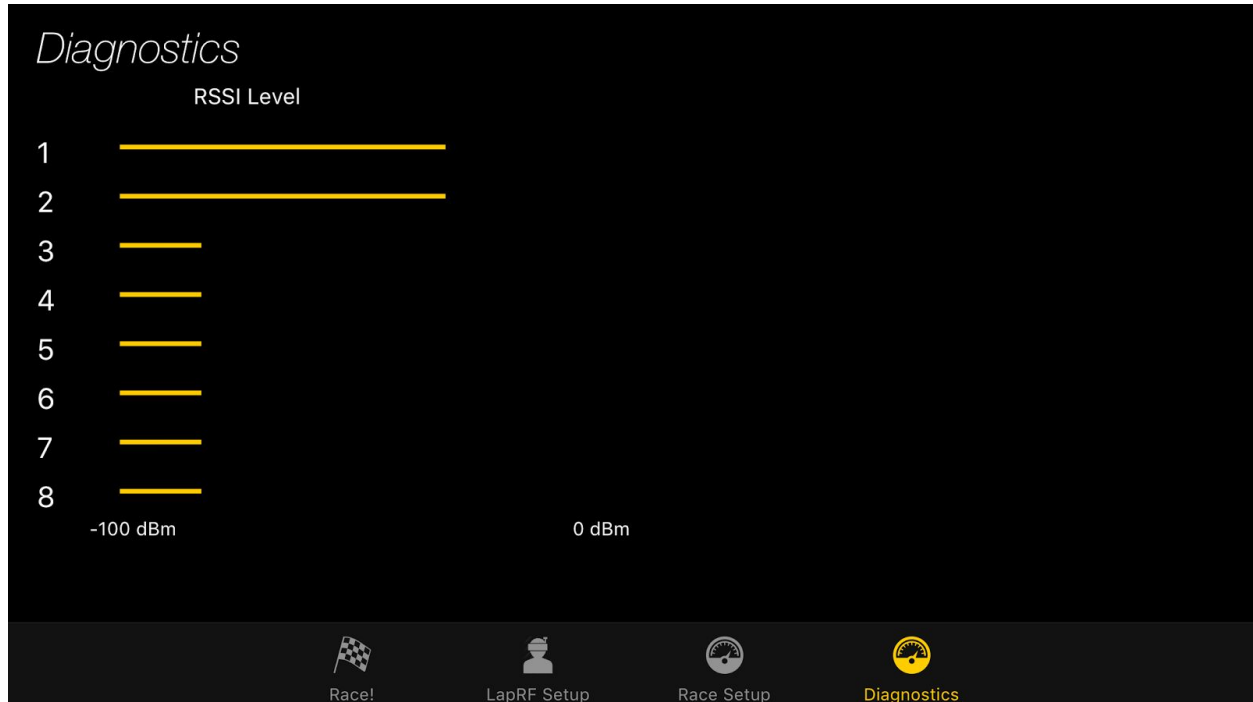
For races which are fixed-time, select 'Fixed Time' and be sure to enter the Race time.

For Practice events, select 'Practice', the race will continue to run and record lap times until your LiPos croak. *Note that only the first 8 times will be shown on the Race page.*

Diagnostics

The diagnostics page shouldn't normally be required, but can be a useful tool during debugging.

The RSSI level of each configured pilot slot is shown. Power on video transmitters and walk them through the gate to ensure correct configuration of the lapRF, and the video transmitter itself.



Bands, Channels, and Frequencies

The applications which support the lapRF define the bands and channels that are available for a race.

All of the commonly used bands are present.

	1	2	3	4	5	6	7	8	
L	5362	5399	5436	5473	5510	5547	5584	5621	Low Band*
F	5740	5760	5780	5800	5820	5840	5860	5880	IRC/FS
R	5658	5695	5732	5769	5806	5843	5880	5917	RaceBand
E	5705	5685	5665	5645	5885	5905	5925	5945	Band E
B	5733	5752	5771	5790	5809	5828	5847	5866	Band B
A	5865	5845	5825	5805	5785	5765	5745	5725	Band A



Note: The band that we refer to as 'Low Band' is not legal in most countries worldwide without an appropriate license. In many countries a special event license may be obtained for official races in areas where interference may be minimized.
Please check with your local radio-communication authority before using this band.

Increasing Bluetooth Range

The internal bluetooth module uses a self-adhesive dipole antenna, fixed to the inside of the product.

This antenna uses a standard U.FL connector, which may be removed, and replaced with an external antenna, by means of a 10cm U.FL to bulkhead SMA cable.

When disassembling the case, a small plastic 'blanking plate' may be removed, to open up the hole through which to secure the SMA connector.

For the antenna, the FatShark 2.4GHz dipole works well:

<http://www.getfpv.com/2-4-ghz-2-5-dbi-antenna-tested.html>

And for the required cable, U.FL to SMA:

<https://www.digikey.com/product-detail/en/amphenol-rf-division/336313-12-0150/ARF2362-ND/5417988>

Note: ImmersionRC will be supplying a kit with the required antenna and cable in the near future, but in the interest of speed to market, the above combination will work well.



Certification

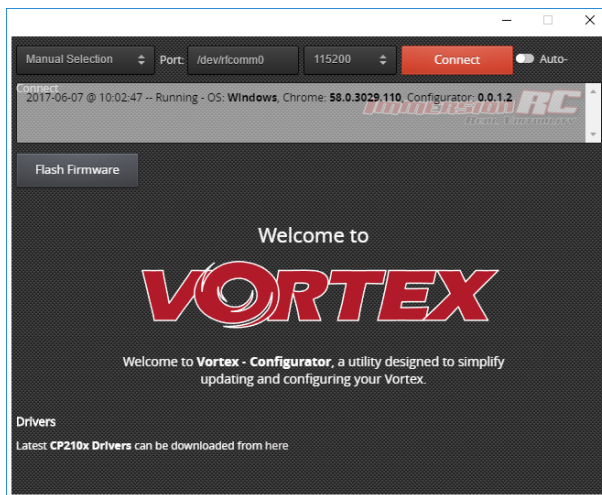
The CE, FCC, and other certificates for the integrated Bluetooth module are available from here:

<https://www.immersionrc.com/fpv-products/laprf/>

Note that these certificates only apply when the internal antenna is used.

Firmware Update

- Install a fully charged battery
- Press and hold the **Bind** button, while pressing the **Power** button
- Run the ImmersionRC 'Vortex Configurator' application
- Press the **Flash Firmware** button, and point to the firmware file downloaded from the ImmersionRC website [product page](#) for the LapRF.



Troubleshooting

Problem: LEDs are white on power-up

Solution: Indicates an issue with the programming of the unique device ID, please contact support.